

David Fernandez

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SUMMARY

I am a PhD candidate in Computer Science at Clemson University, bridging explainable AI, agentic systems, and security for safety-critical applications. My research spans Large Language Models (LLMs), Vision Language Models (VLMs), and deep learning architectures, with five first-authored publications advancing component-level explainability, zero-shot reasoning, and adversarial scenario analysis. As a member of Clemson's VIPR-GS Research Program, I develop hierarchical LLM reasoning frameworks and VLM evaluation systems for the U.S. Army's Next Generation Combat Vehicle (NGCV) program. At BMW Group, I design agentic AI systems for enterprise environments, including autonomous prompt optimization pipelines for continual agent improvement without model retraining, and context-aware moderation frameworks for detecting coordinated multi-turn adversarial attacks in production deployments. I focus on creating robust, interpretable AI that translates cutting-edge research into real-world impact.

EDUCATION

2024 - present	Ph.D. in Computer Science at Clemson University	(GPA: 4.0/4.0)
2022 - 2024	M.S. in Data Science and Informatics at Clemson University	(GPA: 4.0/4.0)
2010 - 2018	B.S. in Computer Engineering at ITAM	(Mexico City, Mexico)
2010 - 2018	B.S. in Business Administration at ITAM	(Mexico City, Mexico)

WORK EXPERIENCE

Virtual Prototyping of Ground Systems (VIPR-GS) for U.S. Army NGCV program, Clemson University 2025 - Present

- Developed perception model architectures using semantic embeddings to bridge visual scene understanding with regulatory compliance, enabling component-level diagnostics in autonomous systems
- Built novel multi-modal fusion architectures combining camera and sensor data, achieving quantifiable improvements in perception accuracy for safety-critical scenarios
- Integrated VLMs with high-fidelity simulations to evaluate multimodal decision-making
- Created MetaVLM evaluation system demonstrating 99.6% performance retention with 43% reduced computational overhead and 9% lower inference latency
- Integrated VLMs with semantic embeddings for zero-shot reasoning, achieving 7% improvement over deep learning baselines

AI Researcher at BMW Group 2024 - Present

- Co-authored SACAIR 2025 publication on edge AI deployment, implementing small language model framework enabling distributed inference on constrained devices
- Designed multilingual semantic document framework using transformer embeddings to automate compliance verification, detecting structural and semantic changes across regulatory documents
- Conducted systematic security analysis of production LLM systems through 10,000+ multi-turn adversarial interactions, discovering critical prompt injection vulnerabilities
- Architected novel context-aware defense framework addressing fundamental limitations in existing industry solutions, researching new security paradigm for enterprise conversational AI systems

PhD Research Assistant at Clemson University 2024 - Present

- Published 5 first-authored papers advancing explainable AI through collaborative research with Army VIPR and BMW, bridging academic work with real-world deployment in safety-critical AI systems
- Teaching Assistant for "Security in Emerging Computing and Networking Systems," delivering lectures, mentoring students on semester-long research projects in adversarial ML and security topics

Lead Engineer & Technical Architect at MUSEOLAB

2016 - 2021

- Designed and deployed 5 production ML pipelines with ensemble methods and clustering algorithms, establishing foundation in model development, optimization, and real-world deployment
- Led full-stack development of ML-integrated systems, including model training, optimization, and production deployment workflows
- Drove agile development practices using Scrum to deliver projects and meet client requirements

AWARDS & HONORS

2026 **SAE International Award for Outstanding Student Paper honoring Phil Myers**

For “Avoiding the Crash: A Vision-Language Model Evaluation of Critical Traffic Scenarios”

PUBLICATIONS

Duffy, Edward B. et al. (2025). “Small Language Models on the Edge for Real-World Agentic Systems in Industry”. In: *Southern African Conference for Artificial Intelligence Research (SACAIR)*. To appear. URL: <https://2025.sacair.org.za/>.

Fernandez, David, Pedram MohajerAnsari, Cigdem Kokenoz, Amir Salarpour, Bing Li, and Mert D. Pesé (2025). “WIP: From Detection to Explanation: Using LLMs for Adversarial Scenario Analysis in Vehicles”. In: *3rd USENIX Symposium on Vehicle Security and Privacy (VehicleSec '25)*. URL: <https://www.usenix.org/conference/vehiclesec25/presentation/fernandez>.

Fernandez, David, Pedram MohajerAnsari, Amir Salarpour, and Mert D. Pesé (2025). “Avoiding the Crash: A Vision-Language Model Evaluation of Critical Traffic Scenarios”. In: DOI: [10.4271/2025-01-8213](https://doi.org/10.4271/2025-01-8213). URL: <https://doi.org/10.4271/2025-01-8213>.

MohajerAnsari, Pedram et al. (2025). “Attention-Aware Temporal Adversarial Shadows on Traffic Sign Sequences”. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, pp. 3600–3608.

Salarpour, Amir, Pedram MohajerAnsari, David Fernandez, and Mert D. Pesé (2025). “SASA: Sequence-Aware Shadow Attacks via Attention Alignment for Traffic Sign Recognition”. In: *The 6th Workshop of Adversarial Machine Learning on Computer Vision: Safety of Vision-Language Agents*. URL: <https://openreview.net/forum?id=UNf8Vo6sPF>.

Fernandez, David, Pedram MohajerAnsari, Cigdem Kokenoz, Amir Salarpour, Bing Li, and Mert D. Pesé (2026). “From MIRAGE to CLEAR: Component-Level Explainable Anomaly Reasoning for Autonomous Vehicle Perception Systems”. In: *International Conference on Dependable Systems and Networks*. To appear.

Fernandez, David, Pedram MohajerAnsari, Amir Salarpour, Long Cheng, et al. (2026). “Comparative Analysis of Patch Attack on VLM-Based Autonomous Driving Architectures”. In: *2026 IEEE Intelligent Vehicles Symposium (IV)*. To appear.

Fernandez, David, Pedram MohajerAnsari, Amir Salarpour, and Mert D. Pesé (2026). “Understanding Adversarial Transferability in Vision-Language Models for Autonomous Driving: A Cross-Architecture Analysis”. In: DOI: [10.4271/2026-01-0170](https://doi.org/10.4271/2026-01-0170). URL: <https://doi.org/10.4271/2026-01-0170>.

Salarpour, Amir, Pedram MohajerAnsari, David Fernandez, Cigdem Kokenoz, et al. (2026). “Trustworthy AI for CPS Security”. In: *AI for CPS Security and Resilience*. Ed. by M. Hadi Amini et al. To appear. Springer.

SKILLS

Programming Languages	Python, Java, Swift, JavaScript, Shell Scripting, HTML
Libraries / Architectures	PyTorch, CUDA, Scikit-learn, Transformers, Vision Transformers, CNNs
Computer Vision & ML	Object Detection, Semantic Segmentation, Multi-Modal Fusion, Model Optimization